

SEMINAR III • WEEK 2

Getting Stuck Is Information

This year you'll run into questions you know cold, questions you half-remember, and questions you get wrong for reasons that have nothing to do with each other. Today we build a routine for all of them.

DAY 1

How Do We Solve Problems?

A five-question routine you'll use all year — in math class and outside it.

BEFORE WE START

What can you do, even when you don't know the answer yet?

Look at a problem that looks unfamiliar. You don't have to solve it. Just notice: what's the first move you could make, even without knowing where it leads? Reread it. Look for numbers. Guess at what's being asked. All of that counts — you already know more than "I don't know" gives you credit for.

THE ROUTINE

The Five Questions

Use this whenever you're stuck — on a math problem, or on anything else.



STOP

What information do I have?

Look for the numbers, facts, definitions, labels, diagrams, or instructions in front of you. Don't calculate yet — first figure out what's actually available.

- numbers
- facts
- definitions
- labels
- diagrams
- tables
- instructions
- context

FIND

What am I trying to determine?

Finish the sentence: "I need to find ____."

Saying it out loud — even just in your head — keeps you from calculating before you actually understand the goal.

CONNECT

What skill or information could help?

You don't need the whole solution yet — just a lead worth following.

- addition
- percentages
- fractions
- ratios
- an equation
- a graph
- a formula
- a definition

TRY

What strategy can I try?

The important word is **try**. A strategy doesn't need to be guaranteed to work before you attempt it.

- write an equation
- draw a diagram
- make a table
- estimate
- eliminate choices
- work backward
- try a simpler example

CHECK

Does my answer make sense?

- Too large? Too small?
- Positive or negative?
- Does the unit make sense?
- Did I answer what was actually asked?

This one matters more and more as the year goes on — don't rush past it.

LET'S WORK ONE TOGETHER

A student earns \$15 per hour and works 6 hours. How much before taxes?

- **STOP:** \$15 per hour, 6 hours.
- **FIND:** the total amount earned.
- **CONNECT:** a rate multiplied by the number of hours.
- **TRY:** $15 \times 6 = 90$.
- **CHECK:** \$15 an hour for six hours should land between \$75 and \$100 — \$90 fits.

THIS ROUTINE ISN'T JUST FOR MATH

A project needs a written response, a visual, and a presentation. It's due Friday. It's currently Wednesday.

- **STOP:** three required parts, two days remaining.
- **FIND:** how to finish on time.
- **CONNECT:** planning and prioritization.
- **TRY:** split the work across Wednesday and Thursday.
- **CHECK:** does the plan leave enough time for all three parts?

BEFORE WE MOVE ON

**Which of the five questions do
you think you most often skip?**

DAY 2

Why Did I Get It Wrong?

Five different reasons an answer can be wrong — and five different fixes.

SAME WRONG ANSWER, DIFFERENT REASONS

Three students all answer the same problem incorrectly.

- **Student A** doesn't know the rule.
- **Student B** knows the rule, but makes a calculation error.
- **Student C** solves correctly, but answers a different question.

Do these three students need the same kind of help? An incorrect answer doesn't automatically mean "I don't know how to do this."

THE FRAMEWORK

Five Reasons an Answer Can Be Wrong

Each one calls for a different next step.

KNOWLEDGE ERROR

You don't currently know or remember the information or skill needed.

Example: you don't remember how negative numbers behave during multiplication.

What helps: review the concept, look at a worked example, ask for an explanation.

PROCESS ERROR

You understand the concept, but didn't correctly complete the steps.

Example: you use the correct formula but substitute values into the wrong places.

What helps: compare your work with a correct example, write out every intermediate step.

EXECUTION ERROR

You knew what to do and used the right process, but made a small mistake carrying it out.

Example: copying 36 as 63, or dropping a negative sign.

What helps: slow down, check your work against an estimate.

COMPREHENSION ERROR

You misunderstood what the question was actually asking.

Example: finding the radius when the question asked for the diameter; missing the word "NOT."

What helps: reread the question, underline the actual task.

STRATEGY ERROR

**You understood the problem,
but chose an approach that
wasn't the best way to solve it.**

Example: solving algebraically for several minutes when substituting answer choices would be faster.

What helps: ask yourself, "is there an easier way?"

ALL FIVE, SIDE BY SIDE

Error Types at a Glance

Knowledge

Process

Execution

Comprehension

Strategy

Ask yourself
Did I know the information or skill I needed?
Did I correctly follow the steps?
Did I make a small mistake doing something I knew how to do?
Did I understand the question correctly?
Did I choose an effective approach?

ONE MORE THING

Errors can overlap. A problem might involve more than one issue.

You're not hunting for a perfect label. You're answering one question: **what should I do differently next time?**

LET'S PRACTICE

Error Detective

Read each one, decide what type of error it is, then check yourself.

ERROR DETECTIVE · 1 OF 5

$6 + 3 \times 4 = ?$

Student answers **36** ($6+3=9$, then $9 \times 4=36$)

Process Error

The calculations were correct, but the order of operations was wrong. Review order of operations and retry.

ERROR DETECTIVE · 2 OF 5

An \$80 jacket is discounted 25%.
Student correctly finds the discount is \$20 and answers **\$20**.

Comprehension Error

The discount was calculated correctly, but the question asked for the sale price, not the discount. Identify exactly what's being asked before answering.

ERROR DETECTIVE · 3 OF 5

$-7 + 12 = ?$

Student: "I don't remember what happens when one number is negative."

Knowledge Error

Review adding positive and negative numbers.

ERROR DETECTIVE · 4 OF 5

48 ÷ 6 = ?

Student correctly picks division, writes **48 ÷ 6 = 6**.

Execution Error

Right operation, wrong arithmetic. Check by multiplying the answer back out.

ERROR DETECTIVE · 5 OF 5

A multiple-choice problem could be solved fast by testing the four choices. A student instead builds a long equation, spends six minutes, and doesn't finish — even though the reasoning is valid the whole way through.

Strategy Error

Look for a faster method before starting a long calculation.

BEFORE WE MOVE ON

What type of error most likely happened here, and what should you do next?

DAY 3

ACT Math Baseline

20–24 questions. Diagnostic, not a grade.

WHAT IS A DIAGNOSTIC?

This baseline finds your starting point. It doesn't produce a grade.

- Attempting an unfamiliar question still gives useful information, even if you get it wrong.
- Don't randomly click answers just to finish.
- A wrong answer you actually attempted tells your teacher more than a blank one.

IF TIME REMAINS

**Think back over the baseline
you just took.**

- What felt familiar?
- What felt unfamiliar?
- Where did you feel most confident?

DAY 4

What Does My Baseline Tell Me?

Not just a score — a real starting point.

SAME SCORE, DIFFERENT NEEDS

Two students can get the exact same score and need completely different support.

A score alone can't tell your teacher what to do next — that's why your results get described a different way.

READING YOUR RESULTS

Not Just a Score

Current Strengths
skills you show consistently

Developing Skills
skills you partially show, or seem to remember

Priority Skills — skills that need real instruction or review

This language comes back all quarter, so it's worth getting familiar with now.

A DISTINCTION WORTH MAKING

Learn to tell these two apart:

| *"I don't know this."*

| *"I knew this, but made a mistake."*

They call for completely different next steps — one needs new learning, the other just needs more care.

DAY 5

Build Your Starting Strategy

Name four things you'll carry into the rest of the quarter.

YOUR WEEK 2 REFLECTION

Name Four Things

1. **One Strength** — something you already do reasonably well
2. **One Priority** — something you want to improve
3. **One Strategy** — something you'll try when you get stuck
4. **One Checking Habit** — something you'll do before submitting an answer

KEEP USING THIS ALL QUARTER

Two Routines, All Year



Knowledge → **Process** → **Execution** → **Comprehension**
→ **Strategy** — the five reasons an answer can be wrong.

Week 3 begins Numbers & Operations — the first full run of this routine on real content.